

SAFE OPERATING PROCEDURE

MIG Welder

LOCATION: Various

TOI-OHOMAI
Institute of Technology

PERSONAL PROTECTIVE EQUIPMENT



Version: 2	Key Contact: George Fisher
Effective Date: 18/09/2020	Speciality/Position: Technician
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Permits required	Confined space ☐ Hot Works ☐ Working at Heights ☐

HAZARDS ASSOCIATED WITH THIS EQUIPMENT

PERMANENT HAZARDS Remove as necessary	List	Control Method	Lock out Yes or No
Electric shock	Power, handpiece and earth clamp	Check for any defects in cords and plugs, do not touch live electrical parts	No
Explosion	Sparks can be produced from the welding which can ignite combustible materials in the surrounding area. Identify your work for combustible materials	Remove all flammables, do not weld where flying sparks can strike flammable material, wear PPE for protection from sparks, watch for fire and keep fire extinguisher nearby, do not weld where the atmosphere contains flammable, dust, liquid	
Hot surfaces	Hot hand pieces and hot work material	Do not touch hot parts bare handed. Allow cooling period before working on equipment	
Sharp objects	From welding process	Wear gloves	
Arc Eye	Arc Eye is caused by UV. Too much ultraviolet (UV) light and infrared rays which can burn eyes and skin	Wear an approved welding helmet fitter with the proper shade of filter lens, wear approved safety glasses with side shields under the helmet, use protective screens or barriers to protect others from flash, and sparks. Warn others not to watch	

		<i>the arc, wear body protection made from flame resistant material</i>	
Hazardous materials	Argon, CO ₂	<i>Read SDS sheet</i>	
Heat transfer	Heat can melt metal	<i>Rings and Jewellery must not be worn</i>	
Heart stopping		<i>Persons with pacemakers must maintain a distance (30cm)</i>	

PURPOSE/PROCESS

PROCEDURES:

DO NOT use this machine unless you have proven to your trainer that you are competent in its safe use and operation

Pre Start Check

1. Check workspace, walkways and work leads to ensure that no slip/trip hazards are present.
2. Check work area is clean and free of grease, oil and flammable material.
3. Ensure the equipment and work is dry to avoid electric shocks.
4. Check gloves, welding gun and work leads are in good condition.
5. Ensure adequate ventilation is present before beginning welding operation.
6. Close the UV curtain or erect safety screens to ensure the protection of others.
7. Faulty equipment must not be used, immediately isolate and report suspect machinery.

Operation

1. Ensure machine is correctly set up for current, voltage, wire feed and gas flow.
2. Ensure work earth cable makes firm contact to provide a good electrical connection.
3. Never leave the welder running unattended.
4. Regularly inspect the welding tip and shield for damage.
5. Be aware that work will be very hot after welding



Close Down

1. Switch off the machine.
2. Close the gas cylinder valve.
3. Hang up the welding gun and welding cables and gun can be hot.
4. Leave the work area in a safe, clean and tidy state.

Routine maintenance

Lock out and/or isolate any equipment before inspection or maintenance

	Name	Title/Position	Signature	Date
Author	George Fisher	Technician		
Reviewed by	Marie Dennis	HSW Manager		
Reviewed by		Health and safety rep		
Manager	Joe Nathan			

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Document Control

Date	Revision number	Modifications	Revised by	Approved by
11/12/2019	1	New template	George Fisher	Marie Dennis
17/03/2020	2	Updated	George Fisher	M Dennis